

II. THE CASTLE OF FINANCE: STOCK-FLOW RELATIONSHIP

We have seen in the last section how production leads to income flows across different social actors. Incomes are partly consumed and partly saved. These savings accumulate to form wealth.

Wealth is a stock (like capital stock). The addition to wealth in the form of savings is a flow (like income)

We will discuss how savings flow across various social actors, and how (financial) wealth accumulates.

The world of finance is characterised by instability. For instance, financial wealth in the form of equity holdings can go up or down in value depending on the price of shares.

We need to see what is on the asset side and the liability side of households, businesses, government, banks and foreign. And how they change over time.

Workers

We assume that all workers' wealth (Ω_w) is held in the form of bank deposits (or currency) (M_w).

Stock: $\Omega_w = M_w$

The workers's savings in a particular year (flow) equals the addition to the stock of workers' wealth. $\dot{M}_w$ is the addition to the stock of bank deposits.

Therefore, Flow: $S_w = \dot{M}_w$

The convention is that all sources of funds (for instance, savings) are given a positive sign, and all uses of funds are given a negative sign.

Therefore, $S_w + (-\dot{M}_w) = 0$

Rentiers

The rentier households hold their wealth (or accumulated savings) in the form of equity ($P_v V$), and bank deposits or currency and (M_r)

Stock relation: $\Omega_r = P_v V + M_r$

So their savings this year are an addition to their bank deposits ($\dot{M}_r$) and the new equities they purchase (amounting to $\dot{V}$, with a value of $P_v \dot{V}$)

Flow relation: $S_r = P_v \dot{V} + \dot{M}_r$

From now on, we shall consider all households, worker and rentier households combined

Businesses

The business's assets are its capital stock (machinery and buildings) as valued by the financial market. The value of the capital stock at replacement value is PK (K is the capital stock, P is its price)

qPK is ‘asset value’ or the valuation put on the capital stock by the financial markets.

On the liability side of business are: domestic and foreign loans, raised equity, and business networth (Ω_b).

Stock relation: $qPK = L_b + eZ_b^* + P_v V + \Omega_b$

Private investment (PI) (or the addition to capital stock) is financed through fresh loans (both domestic and foreign) and fresh equity raised by business firms. It is also financed through the firms’ own savings (retained earnings).

Therefore:

Flow relation: $PI = S_b + P_v \dot{V} + \dot{L}_b + e\dot{Z}_b^*$

Note: New loans ($\dot{L}_b + e\dot{Z}_b^*$) and fresh equity ($\dot{V}$; $P_v \dot{V}$ is the value of this equity)

Banks:

On the asset side of the banking system: the loans given by the banks or the credit issued) to business and government as well as the foreign exchange reserves.

Assets and Liabilities: Central Bank

<i>Liabilities</i>	<i>Assets</i>
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1. Currency with the public (C^P)	1. Credit to the government
2. Reserves: Cash with commercial banks (C^B) & Commercial banks' deposits with the Central Bank	2. International reserves (foreign assets)

The above does not take into account short-term or temporary credit from the Central Bank to commercial banks.

Assets and Liabilities: Commercial Banks

<i>Liabilities</i>	<i>Assets</i>
1. Deposits with commercial banks	1. Credit (or loans) to the private sector and the government
	2. Reserves: Cash with commercial banks (C^B) & Commercial banks' deposits with the Central Bank

Assets and Liabilities: Banking System (Central and Commercial Banks combined)

<i>Liabilities</i>	<i>Assets</i>
1. Currency with the public (C^P)	1. Credit (or loans) to the private sector and the government (L)

2. Deposits (D)	2. International reserves (foreign assets)
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[Reserves appear on the asset side of the commercial bank's balance sheet and on the liability side of the central bank's balance sheet.]

We know that:

$CP + \text{Deposits} = \text{Money supply in the economy (M)}$

Therefore,

Stock relation: $M = L + eR^*$

Against new loans issued and reserves acquired by the banking system ($\dot{L} + e\dot{R}^*$), new money is created.

Flow relation: $S_l + \dot{M} = \dot{L} + e\dot{R}^*$

S_l represents any saving acquired by the banking system. We will assume that $S_l = 0$.

Government

The government's asset is the "full faith and credit" Δ behind its debt obligations, $L_g + eZ^*$

Stock relation: $L_g + eZ^* = \Delta$

The government finances its fiscal deficit (excess of expenditures over debt or negative savings) with fresh loans from domestic and foreign sources.

Flow relation: $S_g = \dot{L}_g + e\dot{Z}_g^*$

Foreign

For foreign entities having economic engagements with “our” domestic economy: the loans we have taken from them (Z^* in dollars, eZ^* in Rupees) are their assets. On the other side of the stock relation are the international reserves held by our banking system (which is effectively credit to foreign) and foreign wealth (on account of their relation with our economy or our external debt (foreign loans) minus international reserves)

$$\text{Stock relation: } eR^* + \Omega_f = eZ^*$$

$$\text{The flow relation: } S_f + e\dot{R}^* = e\dot{Z}^*$$

The Stock relationships: Assets and Liabilities of various social classes

Social classes	Liabilities	Assets
1. Households	Ω_H	$M_H + P_v V$
2. Business	$L_B + P_v V + \Omega_B$	qPK
4.	L_G	Δ
5. Banks	M	L

Social classes	Liabilities	Assets
1. Households	Ω_H	$M_H + P_v V$
2. Business	$L_B + P_v V +$	qPK
4.	$L_G + eZ_G^*$	Δ

5. Banks	M	$L + eR^*$
6. Foreign	$eR^* + \Omega_F$	eZ^*

Summing across all balance sheets:

$$\Omega_H + \Omega_B + \Omega_F = qPK + \Delta$$

The “primary wealth” is made up of the government’s debt plus the capital stock valued at its asset price. The primary wealth constitutes the networths of households (workers, rentiers), businesses, and the rest of the world.

Note: $M = M_H = M_W + M_R$ $Z = Z_B + Z_G$ and $L = L_B + L_G$

Summing across all balance sheets:

$$\Delta + qPK = \Omega_w + \Omega_r + \Omega_b + \Omega_f$$

The “primary wealth” is made up of the government’s debt plus the capital stock valued at its asset price. The primary wealth constitutes the net worths of workers, rentiers, businesses, and the rest of the world.

Note: $M = M_w + M_r$

$Z = Z_b + Z_g$ and $L = L_b + L_g$

The flow relationships: Net increases in Assets and Liabilities of various social classes

Social classes	Net Increase in Liabilities	Net increase in Assets
1. Households	S_H	$P_v \dot{V} + \dot{M}_H$
2. Business	$S_B + P_v \dot{V} + \dot{L}_B + e \dot{Z}^*_B$	PI
3. Banks	$\dot{M}$	$\dot{L} + e \dot{R}^*$
4. Government	(S_g)	$-(\dot{L}_G + e \dot{Z}^*_G)$
5. Foreign	$S_F + e \dot{R}^*$	$e \dot{Z}^*$

$$\dot{M}_w + \dot{M}_r = \dot{M}_H = \dot{M}$$

$$\dot{Z}^*_G + \dot{Z}^*_B = \dot{Z}^*$$

$$\dot{L}_B + \dot{L}_G = \dot{L}$$

S_G is typically negative = fiscal deficit

Adding up the sources and uses of funds and cancelling out items

$$S_H + S_B + S_G + S_F = \text{PI}$$

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